

KAUSHIK BASAVARAJU

Data Science Professional

PROFESSIONAL SUMMARY

Quantitative-minded Data Science professional with a strong foundation in mathematical modeling, machine learning, and statistical analysis. Experienced in developing research-driven, scalable solutions using Python, time-series data, and real-world datasets. Passionate about applying creative data strategies to complex problems in dynamic environments like finance and economics.

EMPLOYMENT HISTORY

ASSOCIATE SOFTWARE ENGINEER EY GDS

Aug 2022 - Jul 2023

- Spearheaded the end-to-end development of a scalable e-commerce platform using React.js integrated with Salesforce B2B Commerce data APIs for real-time product catalogs, pricing, and transaction history.
- Utilized Apex to fetch real-time product data from Salesforce B2B and integrated with React.js frontend for building custom product displays, including features like product addition, pricing integration, category filters, quantity management, payment method selection, and dynamic image uploads.
- Developed functionality to export product data and generate downloadable reports, increasing store usability for administrators.
- Built and deployed a dynamic blog platform from scratch using React.js, HTML, CSS, and Tailwind with WYSIWYG editing, category classification, image embedding, and live content preview.
- Implemented admin functionalities with authentication, edit/delete routes, and seamless routing using react-router-dom, and fetched real-time content from local JSON server.
- Designed visually rich UI using advanced CSS, Google Fonts, and responsive design principles for a seamless cross-device experience.
- Collaborated with QA and backend teams to debug API integration issues, validate real-time rendering logic, and ensure smooth user navigation.
- Used Python scripts for parsing and transforming JSON blog data for backend integrity checks and structured rendering.
- Applied version control using Git/GitHub and deployed in modular environments with automated build tools, following Agile/Scrum practices.

PROJECT INTERN IN DATA SCIENCE Verzeo Edu Tech Pvt Ltd (Co-Sponsored by Microsoft)

Mar 2021 - Apr 2021

- Designed and developed a Resume Screening System as part of a capstone project using Natural Language Processing (NLP) to automate candidate evaluation and shortlisting from unstructured resume documents.
- Built a data ingestion and preprocessing pipeline utilizing Python, Pandas, and NumPy, with capabilities for tokenization, stop-word removal, stemming, lemmatization, and regular expressions to clean and extract textual information.
- Applied Named Entity Recognition (NER) to extract structured entities such as name, contact, skills, education, experience, and certifications from resume PDFs and text files.
- Engineered features using TF-IDF, CountVectorizer, and Word Embeddings (Word2Vec) to represent resume content numerically for downstream analysis.
- Built and fine-tuned classification models using Scikit-learn, including Logistic Regression, Naive Bayes, and Random Forest Classifier, to predict candidate suitability based on job descriptions.
- Evaluated model performance using cross-validation, confusion matrix, accuracy score, precision, recall, and F1-score to ensure robustness and fairness in candidate ranking.
- Developed a Streamlit web app for an interactive user interface where HR managers can upload resumes and receive real-time ranked recommendations along with candidate summary insights.
- Automated backend data integrity checks using Python scripts to validate product JSON structures.
- Practiced Agile/Scrum methodology and used Git/GitHub for version control and modular deployment.
- Contributed to discussions on backend architecture scalability and data consistency.
- Visualized key insights using Matplotlib and Seaborn, including word clouds, top skill frequency, and candidate match scoring for presentation to stakeholders.
- Utilized Git and GitHub for source control, Jupyter Notebook for experimentation, and adhered to Agile methodologies for weekly deliverables and sprints.

- Focused on real-world problem-solving, AI in recruitment, and ethical data practices, contributing to the broader impact of AI-powered hiring solutions.

EDUCATION

MASTER'S, DATA SCIENCE
University of Massachusetts - Dartmouth

Aug 2023 - May 2025

BACHELOR OF TECHNOLOGY, COMPUTER SCIENCE AND TECHNOLOGY
GITAM University

Jun 2018 - Jun 2022

SKILLS

Python (Advanced), SQL (Advanced), JavaScript, HTML5, CSS3, React.js, Flask, Node.js, FastAPI (Familiar), PostgreSQL, MySQL, MongoDB, AWS (EC2, S3, Lambda), Apache Kafka (Basic), Airflow (Learning), Git, GitHub, Docker (Basic), Jenkins (Familiar), ETL Pipelines, Data Modeling, Data Warehousing, Streamlit, Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, pgAdmin, Scikit-learn, Word2Vec, TF-IDF, Random Forest, NLP, LLMs (Flan-T5), Data Visualization, Agile/Scrum, CI/CD, Statistical Analysis,

Quantitative & Research Skills: Statistical Modeling, Time-Series Forecasting, LSTM, Experimental Design, Probability Theory, Signal Processing, Data Denoising, Regression Analysis, Linear Algebra, Feature Engineering,

Probability & Statistics Linear Regression Exploratory Data Analysis Mathematical Modeling Error Metrics: MAE, RMSE, Cross-validation.

LINKS

LinkedIn: [linkedin.com](https://www.linkedin.com), GitHub: github.com.

ADDITIONAL INFORMATION

MOVIE RECOMMENDATION SYSTEM

Tools: Flask, React.js, Pandas, NumPy, REST API, Axios

- ♦ Built a full-stack movie recommendation engine using Flask for the backend and React.js for the frontend, enabling seamless RESTful communication via Axios.
- ♦ Engineered a content-based filtering algorithm using Pandas and NumPy to generate ranked movie suggestions.
- ♦ Designed scalable API routes for search and recommendation, improving API latency by 30% through backend optimization.
- ♦ Developed a dynamic UI for real-time recommendations based on user input and movie attributes.

VOICE-CONTROLLED ROBOT MOVEMENT SIMULATION

Tools: Python, SpeechRecognition, pytsx3, turtle

- ♦ Developed a Python-based simulation for a voice-controlled robot integrating real-time speech recognition and text-to-speech feedback.
- ♦ Engineered a grid-based robot navigation system that converts commands like “move forward” into movement actions.
- ♦ Improved voice command accuracy to 95% by tuning speech model thresholds and optimizing feedback latency.
- ♦ Simulated and visualized robotic behavior using the turtle graphics library, enabling testing of movement logic.

ETL PIPELINE FOR BORDER DATA ANALYSIS

Tools: Python, PostgreSQL, Pandas

- ♦ Designed a robust ETL pipeline to analyze U.S. border crossing data, transforming and loading over 500,000 records into PostgreSQL for querying.
- ♦ Created custom SQL queries to extract trends by port, transportation type, and month to identify traffic patterns.
- ♦ Applied advanced transformation techniques using Pandas and automated the pipeline for repeatable data refresh cycles.
- ♦ Reduced data cleaning and processing time by over 40% by integrating batch-wise transformation strategies.

INTERVIEW QUESTION GENERATOR

Tools: Streamlit, Hugging Face Transformers (T5), Python

- ♦ Developed an AI-powered interview question generator using Google's FLAN-T5 model to generate tailored technical and behavioral questions.
- ♦ Integrated top-k and top-p sampling strategies to improve relevance and diversity of generated content.
- ♦ Designed a user-friendly interface in Streamlit to allow recruiters or students to select topics and question difficulty.
- ♦ Enabled filtering by category, built reusable prompt templates, and deployed the app with performance-tuned settings.

MACROECONOMIC IMPACT MODELING ON S&P 500

Tools: Python, Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Yahoo Finance API, FRED API

- ◆ Collected and integrated historical S&P 500 data with macroeconomic indicators (GDP, CPI, interest rates, unemployment) using Yahoo Finance and FRED APIs.
- ◆ Performed time-series feature engineering using lag values, rolling means, and correlation analysis to capture market trends.
- ◆ Built and evaluated multiple predictive models, including ARIMA, linear regression, and LSTM-based deep learning networks for market index forecasting.
- ◆ Visualized the impact of macroeconomic shocks on S&P 500 performance using Matplotlib and Seaborn for trend and anomaly analysis.
- ◆ Demonstrated model reliability in identifying leading and lagging indicators influencing market behavior, supporting potential decision-making use cases.

FINANCIAL TIME-SERIES FORECASTING PROJECT

Tools: Python, Keras, TensorFlow, LSTM, Matplotlib, Scikit-learn

- ◆ Developed an LSTM-based time-series forecasting model to predict stock prices using historical S&P 500 data
- ◆ Applied sliding window sampling, normalization, and sequence-to-one structure for temporal feature engineering
- ◆ Implemented MAE and RMSE as evaluation metrics and used walk-forward validation to simulate real-world drift
- ◆ Tuned hyperparameters including time lag, dropout, and learning rate to optimize forecasting accuracy
- ◆ Analyzed performance under high volatility conditions and applied smoothing to extract signal from noisy data
- ◆ Demonstrated a research-driven workflow with iterative experimentation and statistical evaluation